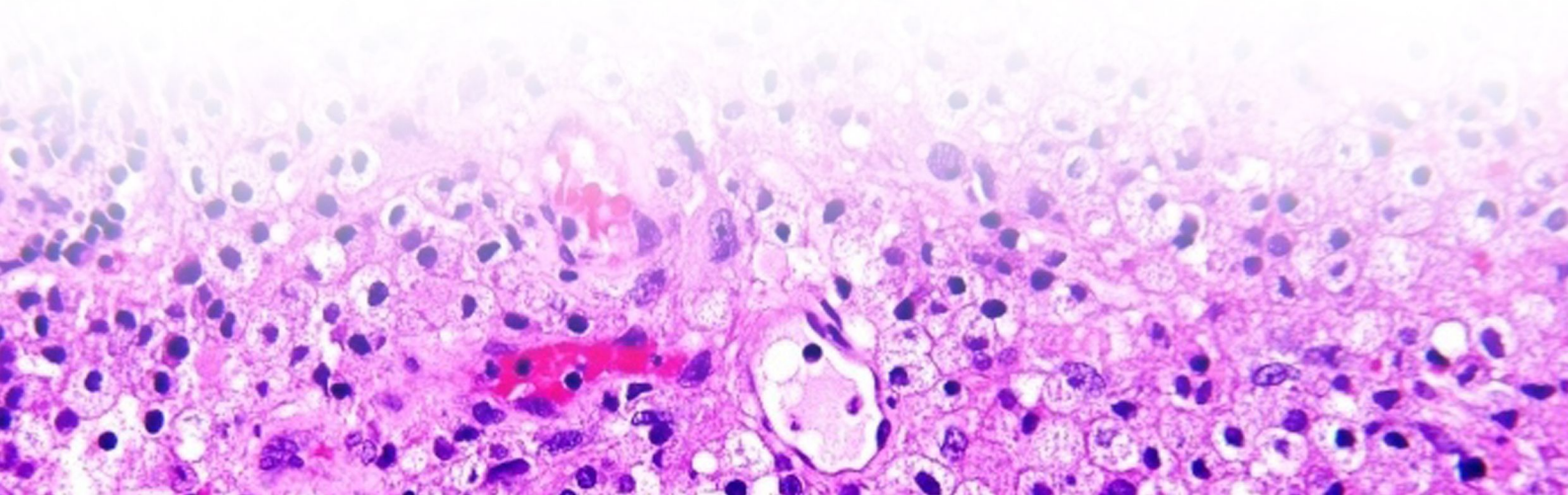
Student Notes

PATHOLOGY

1st Edition

Unit 7

Infectious Diseases



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

- Neelavathi S
- Manivasagam R
- Artheshivani S M
- Kavya M
- Vikram R R
- Subhashree B
- Praveen S P
- Vignesh. M
- Varsha L
- Muthamil Selvi E
- Kaushik N R

INFECTIOUS DISEASES

ESSAY:

1. Define gangrene, types, details about gas gangrene
2. Typhoid ulcer
3. Describe the pathogenesis, morphology and clinical features of TB

SHORT NOTES:

1. Mycetoma
2. Morphology and evolution of tubercle
3. Immunology of leprosy
4. Difference between Tuberculous leprosy and lepromatous leprosy
5. Microscopic appearance of lepromatous leprosy
6. Difference between Dry and Wet gangrene
7. Pathogenesis of gangrene
8. Tertiary Syphilis
9. Viral Hemorrhagic Fever
10. Actinomycosis

SHORT ANSWERS:

1. Cysticercosis
2. Hematocrit in Dengue fever
3. Black Water Fever
4. Gross Appearance of Mycetoma
5. Fungal granuloma
6. Rhinosporidiosis
7. Negri Bodies

UPDATES

ESSAY

1. GANGRENE:

A form of necrosis of tissues with superadded putrefaction. Here the necrosis undergoes liquefaction by the action of Putrefactive Bacteria.

TYPES:

- Dry gangrene.
- Wet gangrene
- Gas gangrene(a type of wet gangrene)

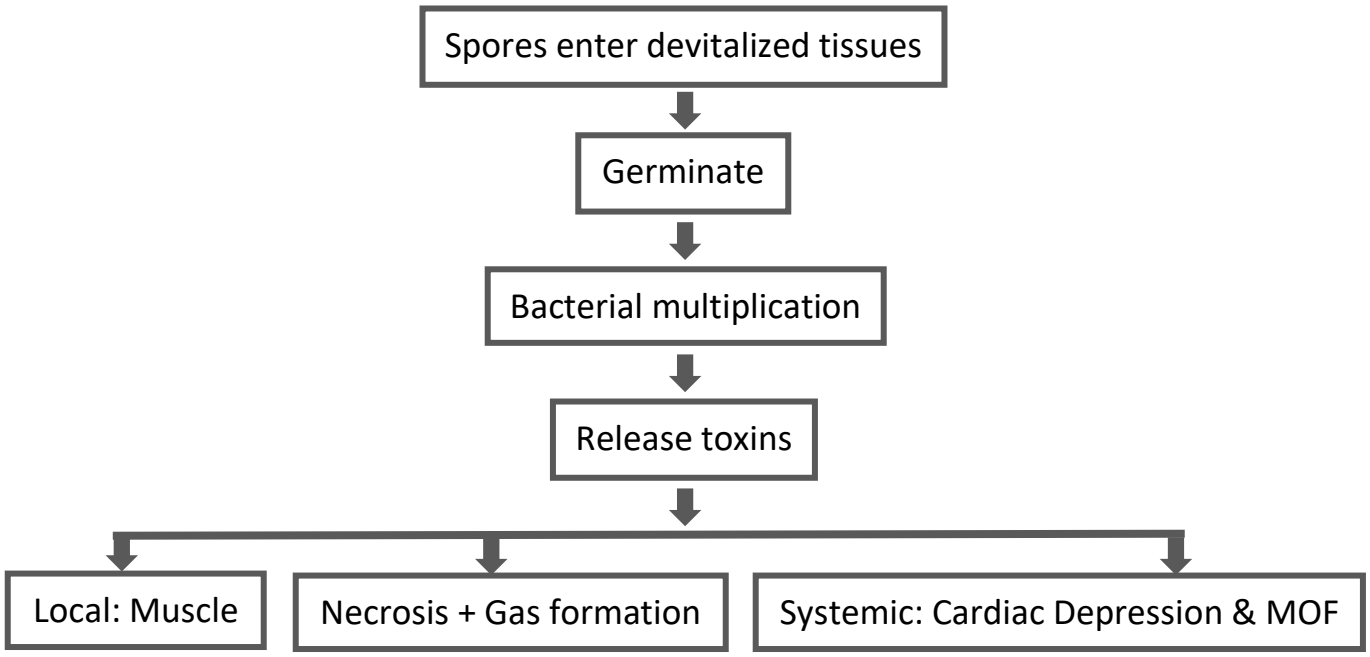
GAS GANGRENE:

Deep wound infection, destructive of tissues caused by combination of 2 or more species.

CAUSATIVE AGENT:

- Clostridium perfringens
- Clostridium novyi
- Clostridium septicum

PATHOGENESIS:



MORPHOLOGY:

GROSS:

- Affected area is swollen, oedematous, painful and crepitant due to accumulation of gas bubbles of CO2
- Subsequently, the affected tissue become Dark Black and foul smelling.

MICROSCOPY:

- Muscle fibres undergo Coagulative necrosis with Liquefaction.
- Large number of Gram +ve bacilli are seen.
- At Periphery, a leucocytic infiltration, Oedema and Congestion are found.
- Capillary and Venous thrombi are common.

SYMPTOMS:

- High fever
- Shock
- Massive tissue destruction
- Blackening of skin
- Blisters with gas bubbles from infected area.

TREATMENT:

- High dose of antibiotic – Penicillin
- Dead tissue is removed / Limbs amputated
- No vaccine.

2. TYPHOID ULCER:

CAUSATIVE AGENT:

- Salmonella typhimurium
- Salmonella paratyphi

PATHOGENESIS:

- Salmonella possesses virulence genes that encode a type3 secretion system capable of transferring bacterial proteins into the enterocytes.
- Transferred proteins activate host cell Rho GTPases, bacterial uptake that allow bacterial growth within phagosomes.
- Flagellin, the core protein of bacterial flagella, activates TLR5 on host cell and increase local inflammatory response.

MORPHOLOGY:

- Affects ileum and colon.
- Present as longitudinal ulcers over the Peyer's patches.
- Neutrophil accumulate within the superficial layers & macrophages containing bacteria, RBCs, Nuclear debris mix with lymphocytes & Plasma cells in LP.
- Mucosal shedding create oval longitudinal ulcers.
- The Base of Ulcer is Black due to the Sloughed mucosa.
- The Margins of the Ulcers are slightly raised due to inflammatory oedema and Cellular Proliferation.
- No Significant Fibrosis.
- Fibrous stenosis seldom occur in healed Typhoid Lesions.
- Regional Lymph nodes are invariably enlarged.
- Blunting of villi.
- Mucosal congestion & edema.

CLINICAL FEATURES:

- Anorexia, Abdominal pain, Bloating, Nausea, Vomiting & Bloody diarrhoea.
- Blood cultures are positive.
- In chronic phase- Rose spots are seen on chest & abdomen.
- Extra intestinal complications.

3. TUBERCULOSIS

A serious, communicable chronic granulomatous pulmonary & systemic disease caused by Mycobacterium tuberculosis.

PATHOGENESIS:

The outcome of infection depends on development of anti-mycobacterial T-Cell mediated immunity.

- **Entry into macrophages**: By binding to Mannose binding lectin receptor & CR3.
- **Replication in macrophages:** Blocks Phagolysosome formation by inhibiting Ca2+ signals.
- **TH1 response:** Activated by IL-12 produced by action of mycobacterial ligand on TLR-2.
- **TH1 mediated macrophage activation and killing of bacteria:** IFN Gamma expose Bacteria to acidic environment, Express iNOS, mobilize anti-microbial peptide (Defensin).
- **Granulomatous inflammation and Tissue damage:** TH1 response orchestrates the formation of granuloma and caseous necrosis.
- **Role of other immune cells:** NKT cells recognize mycobacteria lipid antigens bound to CD1 produce IFN-G.
- **Host susceptibility to infections:**
  - Polymorphism in IL-12,
  - IFN – Gamma & its receptor
  - HLA

  } Vulnerable state
- **Immunological state in active TB:** Neutrophils of active TB patients have upregulated Type 1 & 2 IFN gene.

MORPHOLOGY:

PRIMARY TUBERCULOSIS:

- Caseating and noncaseating tubercles are seen microscopically
- The granulomas are usually enclosed within a fibroblastic rim punctuated by lymphocytes.
- Multinucleated giant cells are present in the granulomas.

Inhaled bacilli implant in the distal airspaces of lower part of the upper lobe or upper part of lower lobe close to pleura

↓

Sensitization develops

↓

1-1.5cm are of grey white inflammation with consolidation emerges – **Ghon's focus**

↓

Center undergoes caseous necrosis

↓

This combination of parenchymal lung lesion and lymph nodal development is **Ghon's complex**

↓

In 95% cases, cell mediated immunity controls infection

↓

In remaining cases, Progressive fibrosis of Ghon's complex

↓

Radiologically detectable calcification – **Ranke's complex**

↓

Seed into organs without any lesions

SECONDARY TUBERCULOSIS:

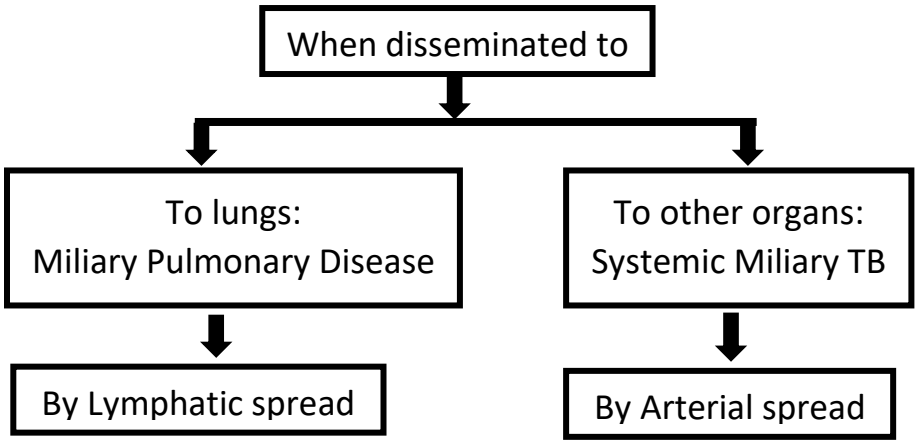
- The initial lesion is usually a small focus of consolidation, less than 2 cm in diameter.
- Foci are sharply circumscribed, firm and gray-white to yellow in color, and have variable degrees of central caseation and peripheral fibrosis.
- Active lesions show characteristic coalescent tubercles with central caseation.

PROGRESSIVE PULMONARY TUBERCULOSIS:

- In older adults and immunosuppressed people.
- Expand to adjacent lung and erodes Bronchi and vessels.
- Erosion of Blood Vessels result in Haemoptysis.

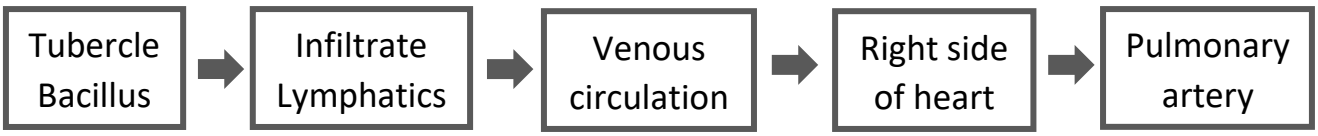
MILIARY TUBERCULOSIS: *(Can be asked in Short Notes)*

Disseminated form of Tuberculosis



A. MILIARY PULMONARY DISEASE:

MODE OF DEVELOPMENT:



MORPHOLOGY:

Multiple, small, yellow, nodular lesion in lung parenchyma as millets so called **Miliary**.

B. SYSTEMIC MILIARY TUBERCULOSIS:

MORPHOLOGY:

Mostly involved organs:

- Liver
- Spleen
- Bone marrow
- Adrenals
- Kidney
- Meninges
- Fallopian tubes
- Vertebra
- Epididymis

ENDOBRONCHIAL, ENDOTRACHEAL AND LARYNGEAL TUBERCULOSIS:

- May develop by spread through lymphatic channels or from expectorated infectious material.
- The mucosal lining may be studded with minute granulomatous lesions that may only be apparent microscopically.

ISOLATED TUBERCULOSIS:

- May appear in any of the organs or tissues seeded hematogenously and may be the presenting manifestation.
- Organs that are commonly involved include the:
  - Meninges (tuberculous meningitis),
  - Kidneys (renal tuberculosis),
  - Adrenals (formerly an important cause of Addison disease),
  - Bones (osteomyelitis), and
  - Fallopian tubes (salpingitis).
- When the vertebrae are affected, the disease is referred to as Pott disease
- Para spinal "cold" abscesses in these patients may track along tissue planes and present as an abdominal or pelvic mass.

LYMPHADENITIS:

- Most frequent presentation of extra-pulmonary tuberculosis, usually occurring in the cervical region ("scrofula").
- In HIV-negative individuals, lymphadenitis tends to be unifocal and localized.
- In HIV-positive people, on the other hand, almost always have multifocal disease, systemic symptoms, and either pulmonary or other organ involvement by active tuberculosis.

CLINICAL FEATURES:

- **Primary TB**: Occurs in previously unexposed unsensitized individuals. Acute bacterial Pneumonia, Hilar adenopathy and Pleural effusion.
- **Secondary TB**: Arise in previously sensitized host. Due to Reactivation of latent infection or Reinfection.
- **Secondary Pulmonary TB**: Cavity formation in lungs which erodes into airways, Malaise, Anorexia, Weight loss & Fever, Night sweats and Pleuritic pain.
- **TB and HIV**: Low CD4+ count leads to TB. There is Focal lesion to multifocal infiltrates to Localized apical disease with cavitation.

DIAGNOSIS:

- Acid-fast smears and cultures of the sputum.
- Comprehensive antibiotic susceptibility testing
- PCR
- **IFN-γ release assays (IGRAs)** - Production of IFN-γ by the T cells is measured to assess the level of T-cell immunity to the organism.
- **The Tuberculin test** (purified protein derivative [PPD] - The tuberculin skin test is performed by intracutaneous injection of purified protein derivative of M. tuberculosis.
- **Mantoux skin test:**
  - **False-negative reactions** - certain viral infections, sarcoidosis, malnutrition, Hodgkin lymphoma, immunosuppression, and (notably) active tuberculous disease.
  - **False-positive tuberculin** - Infection by atypical mycobacteria or prior vaccination with BCG

TREATMENT:

- Antiretroviral therapy (ART)
- Immunosuppressive therapy
- Antibiotics

SHORT NOTES

1. MYCETOMA:

- CHRONIC SUPPURATIVE INFECTION
- LIMBS ARE USUALLY INVOLVED

CHARACTERISTICS:

Sinus → material discharge is form of grain with colonies of fungi and bacteria

2 MAIN TYPES:

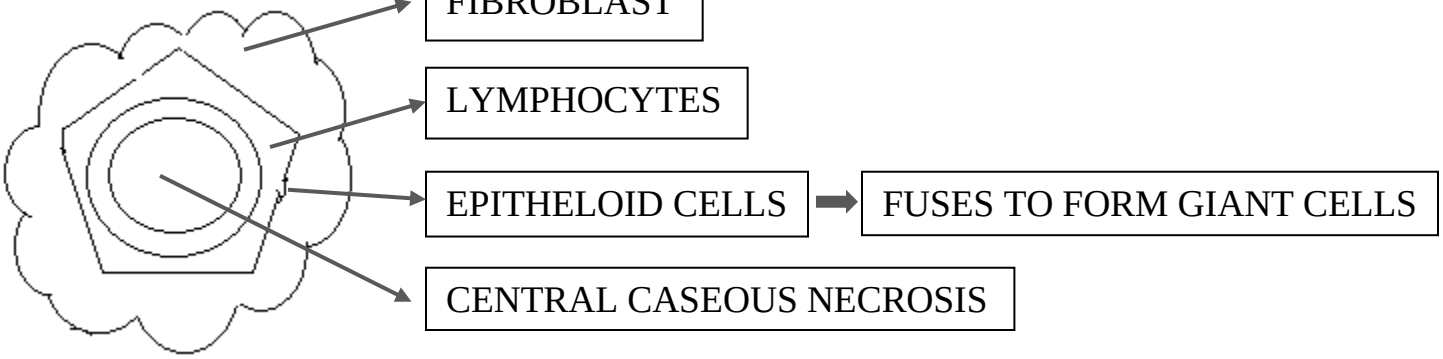
- Mycetoma caused by actinomyces →Actinomycetoma (60% cases)
- Eumycetoma caused by true fungi → Madurella mycetomatis / M.grisea

MORPHOLOGY:

- Foot: Madura foot [swollen foot]
- Penetrate subcutaneous tissue and fascia till bone and draining sinus discharge purulent material and black grai

2. MORPHOLOGY AND EVOLUTION OF TUBERCLE:

MORPHOLOGY:



EVOLUTION:

After 3 weeks of exposure , there is a tubercle development

Antigen presenting cells

Processes mycobacterial Ag

Goes to regional lymph

$CD4^+$ T cell → TH_1 cell

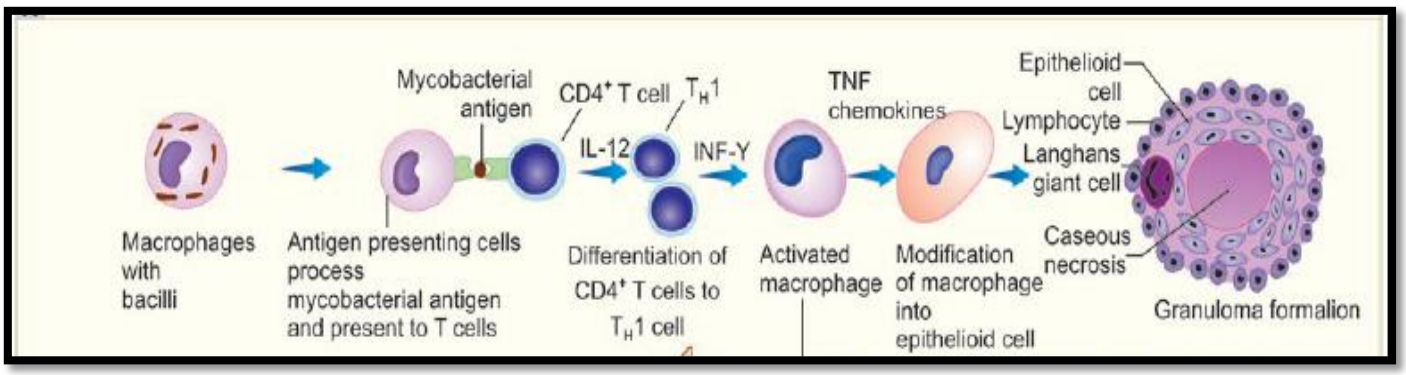
IL-12

MACROPHAGES

INF --GAMMA

MAINROLE IN GRANULOMA FORMATION

- **MACROPHAGE ACTIVATION** AND
- SOME FUSES TO FORM **LANGERHAN GIANT CELLS**
- THIS ARRANGEMENT LEADS TO FORM GRANULOMA ACTIVATION

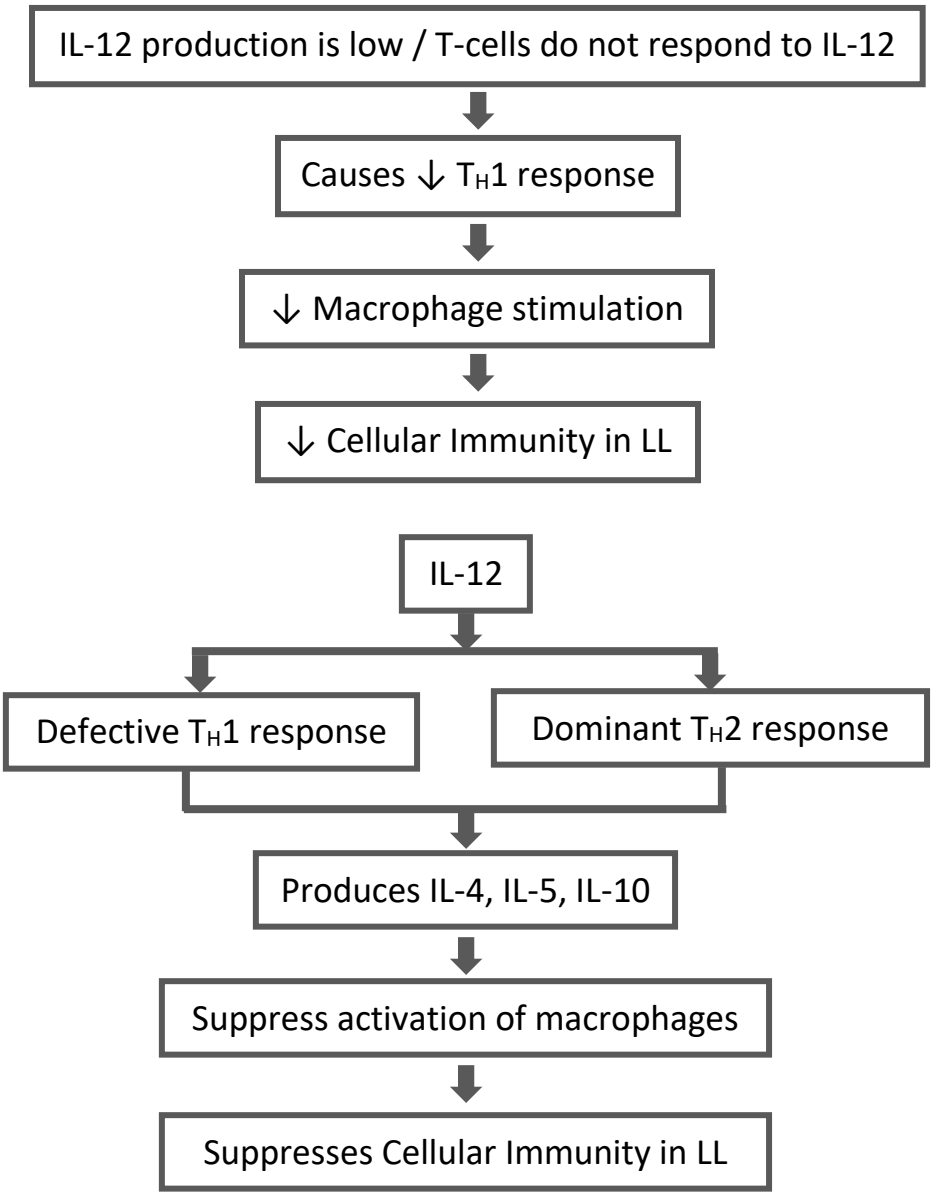


3. IMMUNOLOGY OF LEPROSY:

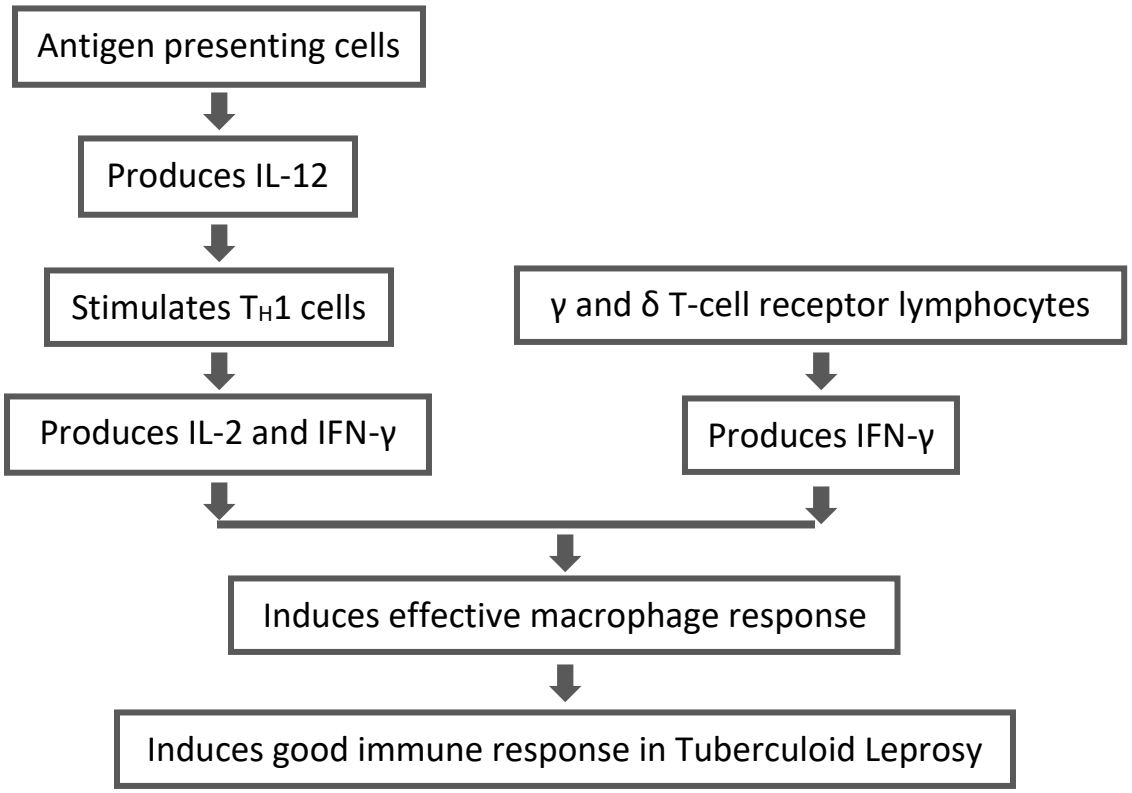
TYPES OF LEPROSY

- **Tuberculoid leprosy** TT
- **Borderline leprosy**
  - Borderline tuberculoid BT,
  - Borderline borderline BB
  - Borderline lepromatous BL
- **Lepromatous leprosy** LL

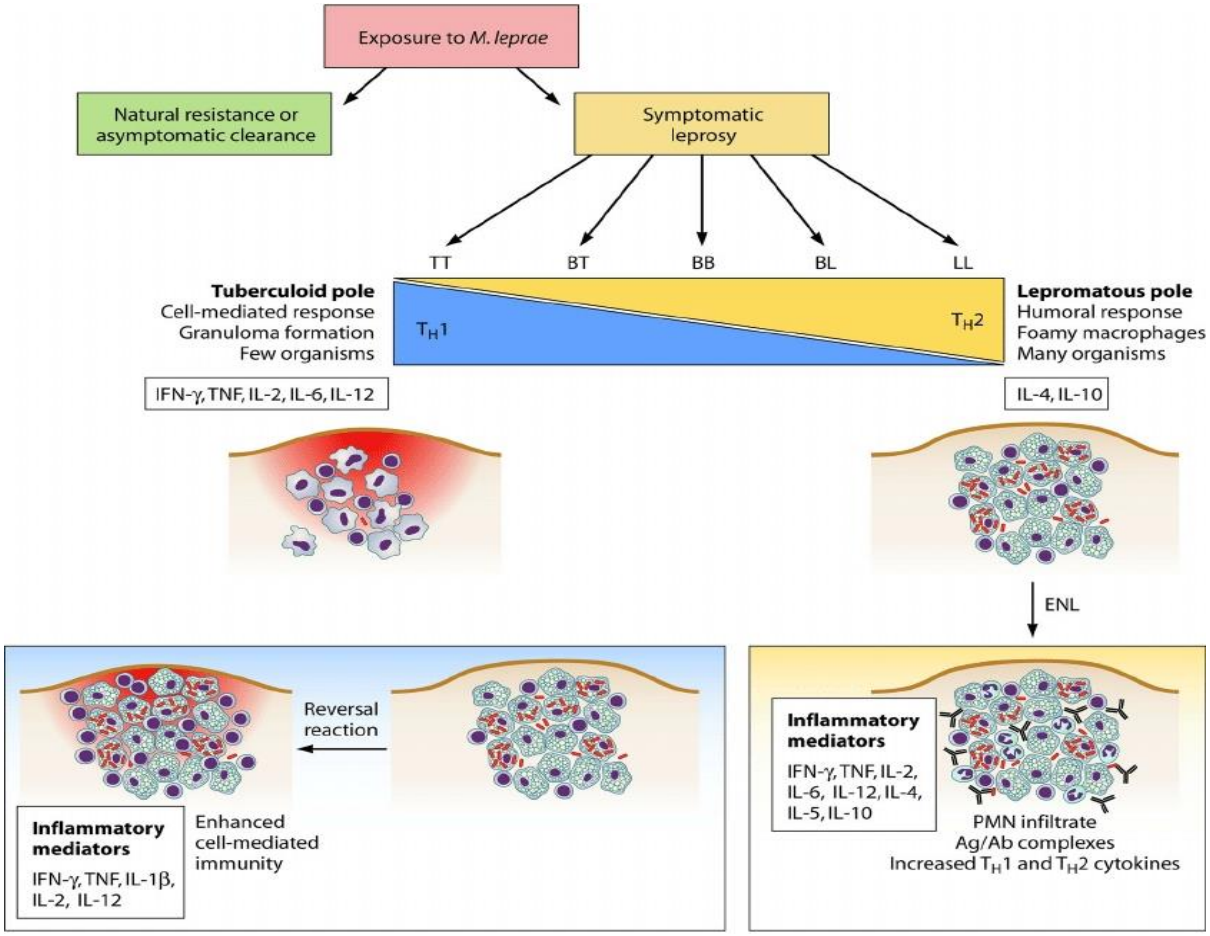
IMMUNOLOGY OF LEPROMATOUS LEPROSY:



IMMUNOLOGY OF TUBERCULOID LEPROSY



IMMUNOLOGY OF LEPROSY:



4. DIFFERENCE BETWEEN TUBERCULOUS AND LEPROMATOUS LEPROSY:

GROSS FEATURES	TUBERCULOUS LEPROSY	LEPROMATOUS LEPROSY
Skin lesions	One or few, asymmetrical Margin is sharp Lesion appear as: • Hypopigmented, annular macules with elevated borders • Dry, scaly Tendency towards central clearing	Many symmetrical Margin is irregular Lesion appear as: • Multiple nodules (lepromata) • Plaques and xantoma-like papules Leonine facies and eyebrow alopecia seen Mostly skin lesions are anesthetic or hypo-esthetic
Nerve lesions	• Early anaesthetic skin lesion • Enlarged thickened nerves • Nerve abscess seen • Asymmetric involvement of large peripheral nerves with granulomatous lesions. • Facial nerve involvement eyelid paralysis, keratitis and corneal ulcer. • Nerve degeneration → skin anesthesia → trauma → chronic skin ulcers. • Contractures, paralysis, auto amputation of fingers and toes occur.	• Nerve lesion appear late • Hypoesthesia is a late sign • Symmetric involvement of nerves without inflammation • Mostly ulnar and peroneal nerves are involved. • Trophic changes follow nerve lesions in hands and feet.
Inflammation	Granulomatous inflammation with few langhans cells	• No inflammation, Virchow's lepra cells-foamy macrophages are more in cigar shaped bundles called globi. • Lymph nodes – have bacteria filled foamy macrophages in Paracortical and Reactive germinal centres. • Macrophages are also seen in splenic red pulp.

GROSS FEATURES	TUBERCULOUS LEPROSY	LEPROMATOUS LEPROSY
Bacillary Load	Paucibacillary	Multibacillary
Bacteriological Index	0-1+	4-6+
Cell-mediated Immunity (CMI)	CMI normal	CMI low
Lepromin test	Positive	Negative
Lymphocyte transformation test	Positive	Negative
CD4/CD8 T-cell ratio	2:1 (normal)	1:2
Humoral immunity	Normal	Exaggerated
Autoantibodies	Not seen	Elevated
VDRL test	VDRL test negative	Biological false positive
Antibodies to phenolic glycolipid (PGL-1)	Elevated in 60% cases	Elevated in 95% cases
Macrophages	Epithelioid type	Foamy type (lipid-laden)
Langerhans giant cells	Found	Not seen
Grenz /clear zone between inflammatory cells and epidermis	Absent	Present
Systemic involvement	Mostly not involved	Mostly involved especially eyes, testes, bones and joints

5. Microscopic appearance of lepromatous leprosy:

- Abundant bacteria due to weak cell mediated immunity and inability to control bacteria → readily visualized in tissue sections **("multibacillary")**
- In smears made from skin lesions, bacteria appear in groups, called cigar-like bundles **(globi)** of weakly acid fast bacilli present inside lipid laden macrophages called **Virchow's lepra cells.**
- The macrophages are able to phagocytose the lepra bacilli but instead of being killed, the bacilli proliferate inside the cells.
- **Flattened epidermis:** Epidermis is thinned and flattened (loss of rete ridges) over the nodules.
- **Grenz (clear) zone:** It is a characteristic narrow, uninvolved dermis (normal collagen) which separates the epidermis from nodular accumulations of macrophages.

6. DIFFERENCE BETWEEN DRY AND WET GANGRENE

DRY GANGRENE	WET GANGRENE
A type of gangrene , which develops in the presence of arterial obstruction , producing sharply localized dryness of dead tissue	The type of gangrene caused by a bacterial infection in the affected tissue
Mechanism : arterial occlusion	Mechanism : venous obstruction
Site : Commonly limbs	Site : More Common in Bowel
Characteristics : dryness , shrinkage , and the black color of the dead tissue , sharply demarcated from adjacent tissue by a line of inflammation	Characteristics : • Softening • Swelling • Blistering • Rotten • Wet and dark appearance
Putrefaction is limited due to very little blood supply	Putrefaction is marked due to the congestion of organs with blood
A line of demarcation is present at the junction between healthy and gangrenous parts	There is no clear cut line between the healthy and gangrenous parts
Bacteria fail to survive	Numerous bacteria are present
Can be treated with surgery	Requires immediate treatment as it can cause death in a few days
Prognosis is better due to little septicemia	Prognosis is poor due to profound toxemia

7. PATHOGENESIS OF GANGRENE

- Clostridium enters the injured tissue
- Spore produced
- Available in circulation
- Available in anaerobic tissue
- Grow and ferment muscle carbohydrate
- Toxins and enzymes produced
- **Enzymes** :
  - Lecithinase
  - Collagenase
  - Permease ,
  - Hyaluronidase
- Enzymes digest and kill host cell and extend anaerobic environment
- Produce gases and destroy tissues (crepitant tissue)

8. TERTIARY SYPHILIS

- About 2-3 years following first exposure , tertiary lesions of syphilis occur
- Less infective

TYPES OF 3° SYPHILIS:

- **Syphilic gumma**:
  - Solitary , localized , rubbery lesion with central necrosis
  - Seen in liver , testis , bone , brain
  - Liver – scarring of hepatic parenchyma (hepar lobatum)
- **Diffuse lesions of tertiary syphilis**:
  - Lesions appear following widespread dissemination of spirochetes in the body

A. CARDIOVASCULAR SYPHILIS

- Also Known as **Syphilitic aortitis**.
- involves thoracic aorta
- wall of aorta weakened – aortic aneurysm
- incompetence of aortic valve

B. NEUROSYPHILIS:

May be Asymptomatic or Symptomatic

Asymptomatic:

Detected by CSF Examination

- Elevated Protein Levels
- **Pleocytosis** (Increased number of inflammatory cells)

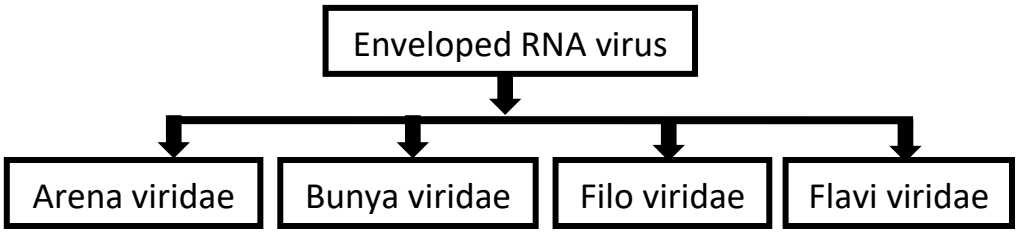
Symptomatic:

- **Meningovascular syphilis** – affects meninges
- **Tabes dorsalis** affecting the spinal cord
- **General paresis** affecting the brain

9. VIRAL HEMORRHAGIC FEVER:

Life-threatening multisystem syndrome + vascular dysregulation and damage → shock.

CAUSES:



SYMPTOMS:

- **Mild Acute Disease** → **Severe life-threatening disease**
  - Fever
  - Headache
  - Myalgia
  - Rash
  - Thrombocytopenia
  - Neutropenia
  - Sudden hemodynamic deterioration
  - shock

TRANSMISSION:

- Pass through animal/ insect host
- Humans → incidental host, infected when come in contact with infected animal [rodents] or insect [rodents, ticks]
- Ebola, Marburg, Lassa → cause haemorrhagic fever → transmitted from person to person.

PATHOGENESIS:

- Hemorrhagic manifestation due to:
  - Hepatic injury → Deficiency of clotting factors
  - Cytokine induced DIC
  - Platelet dysfunction
  - Endothelial injury
  - Thrombocytopenia

} Hemorrhagic manifestations (like petechiae)

- Damage to Blood vessels due to
  - Injury and damage t endothelial cells
  - Infection of dendritic cells and macrophages → Production of inflammatory cytokines
- Hemorrhages → prominent in Congo-Crimean fever → rarely life threatening.
- Necrosis of tissues → secondary to vascular lesions and hemorrhages → varies from mild, focal to massive → attendant inflammatory response minimal.

10. ACTINOMYCOSIS

- Chronic suppurative disease
- Caused by ***actinomycetes (A.israelii)***
- Infection is always endogenous in origin (not by person to person contact)

MORPHOLOGIC FEATURES:

- **Oral cervicofacial actinomycosis**:
  - Commonest form (60%) – best prognosis.
  - Infection enters from tonsils , carious teeth , periodontal disease
- **Thoracic actinomycosis :** infection in the lungs is due to aspiration of the organism from oral cavity or extension of infection from abdominal or hepatic lesions
- **Abdominal actinomycosis :** presents as an abscess , commonly in appendix , caecum , liver
- **Pelvic actinomycosis** : infection in the pelvis occurs most often as a complication of intrauterine contraceptive devices and begins as endometriosis

MICROSCOPIC EXAMINATION:

- Inflammatory reaction is granuloma with central suppuration
- Sulphur granule

SHORT ANSWERS

1. CYSTICERCOSIS

- Caused by ***Taenia solium*.**
- Eggs ingested through contaminated food and water, hatch into larvae, spread hematogenously, and encyst in many organs.
- Encystment in brain- **Neurocysticercosis**.
- Convulsions, increased intracranial pressure and other neurological disturbances occur.
- Most common locations are brain, muscle, heart and skin.
- They may be intraparenchymal, attached to arachnoid or freely floating in ventricles.
- Focal scarring and calcification seen by radiography when cyst degenerates.

2. HAEMATOCRIT IN DENGUE FEVER:

- Haematocrit Elevated due to Haemoconcentration
- Greater than 20%- sign of haemoconcentration and precedes shock.
- Monitored every 24 hours to facilitate early recognition of dengue haemorrhagic fever and every 3-4 hours in severe cases- dengue shock syndrome.

3. BLACK WATER FEVER:

- Also called **malarial hemoglobinuria**.
- Caused by **Plasmodium falciparum**.
- Haemolysis of red cells.
- Rapid pulse, high fever and chills, haemolytic anaemia.
- Urine is black or dark red in colour due to presence of haemoglobin urine.

4. GROSS APPEARANCE OF MYCETOMA:

- Chronic granulomatous disease affecting subcutaneous tissue of the feet and hands.
- Caused by Madurella, Nocardia, Actinomadura and Streptomyces somaliensis.
- Features are:
  - Subcutaneous nodular swelling.
  - Multiple sinuses.
  - Discharge contains granules.
  - Bony deformities.

5. FUNGAL GRANULOMA (Fungus ball)

- A fungus, which is the antigen here, is surrounded by lymphocytes with mast cell .This arrangement is called fungal granuloma.
- The granuloma
  - Has a sharp demarcated borders, discrete spherical appearance.
  - Cavitation is present.

6. RHINOSPORIDOSIS

- A chronic granulomatous disease , where friable polyps developed.
- **Caused by:** Rhinosporidium seeberi(aquatic protozoa)
- **Common presentation:** U/L or B/L nasal polyps.
- **Pathogenesis:** present as polypoidal growth in the nasal cavity, lesions can also be seen in eye, skin, buccal cavity or genitalia.
- Mostly it is mucous membrane. Haematogenous dissemination is rare
- **Diagnosis:** Polyp biopsy done, we get a spherule with many endospores.
- **Treatment:** surgical polypectomy (removal of polyp)

7. NEGRI BODIES

- These are inclusion bodies which are located intracytoplasmic
- Virus particles get replicated in the cytoplasm /nucleus, so during that they form aggregation.
- These negri bodies are characteristic feature of RABIES VIRUS.
- This negri bodies are normally found in hippocampus.

SELLER'S TECHNIQUE

- Negri bodies is detected by Sellers technique.
- **Specimen**: Impression smears of brain stained by sellers technique
- **Stain:** Basic fuschin and methylene blue in methanol.
- Fixation and staining are done simultaneously which is an advantage here

UPDATES:

1. PATHOGENESIS OF GI DISEASE (ADDITIONAL)

- **Toxin production.**
  - Some organisms contaminating food can produce gastrointestinal disease without necessarily establishing an infection in the host.
  - An example is Staphylococcus aureus, which elaborates a powerful exotoxin during its growth in contaminated food that is responsible for acute food poisoning.
- **Bacterial colonoization and toxin production.**
  - Other bacteria establish an infection and produce damaging toxins.
  - Examples include V. cholerae and enterotoxigenic Escherichia coli, which bind to the intestinal epithelium and multiply in the overlying mucous layer.
  - These organisms elaborate potent exotoxins that are responsible for symptomatic disease.

2. ROLE OF CELL-SURFACE RECEPTORS IN MEASLES:

- **Signaling lymphocytic activation molecule family member 1 (SLAMF1)** is expressed on activated lymphocytes, dendritic cells, and monocytes, and it serves as the initial receptor for viral infection.
- **Nectin-4** is found on the basal surface of epithelial cells and is thought to be important for replication of the virus within the respiratory tract, before spread of the virus in respiratory secretions.
- **CD46** was the first cell-surface receptor identified for measles virus, but it has been found to be used only by culture-adapted virus (including the vaccine strain), and not wild-type virus

3. EVASION FROM HOST DEFENCE – LEISHMANIASIS:

To escape killing by neutrophils, Leishmania spp. use the following mechanisms:

- Interfere with the formation of phagolysosomes and fusion with granules
- Localize to nonlytic compartments
- Resist toxicity by reactive oxygen species
- Resist neutrophil extracellular trap (NET) formation by producing endonucleases that digest the NETs and expressing protease-resistant molecules.

4. FORMS OF LEISHMANIASIS:

- **Old World**
  - L. major, L. tropica, L. donovani, and L. infantum
  - Refers to the Eastern Hemisphere (parts of Asia, the Middle East, Africa (particularly the tropical region and North Africa), and southern Europe.
- **New World**
  - L. mexicana, L. braziliensis, and L. chagasi
  - Refers to the Western Hemisphere (parts of Mexico, Central America, and South America).

5. REQUIREMENT OF IRON - LEISHMANIASIS

- Leishmania spp. require iron for survival.
- They have an iron transporter from the ZRT-, IRT-like (ZIP) family of membrane proteins that is expressed on amastigotes and
- The iron transporter stimulates iron entry into the normally low iron environment of macrophage phagolysosomes.
- The host combats this iron acquisition by using pro-inflammatory cytokines to repress iron absorption by
  - Increasing production of hepcidin - the principal iron exporter
  - Activating synthesis of ferritin - binds free iron
- In addition, macrophages downregulate the transferrin receptor and remove iron from the phagosome.

6. CMV INFECTION

- Adoptive transfer of CMV-specific T cells has been successfully used to prevent CMV disease after bone marrow transplantation, and the CD8+ T cell response has been thought to be the most important effector response.
- CD4+ T cells, γδ T cells, and NK cells are known to play a role in immune control of the infection.
- CMV encodes several Fcγ-binding glycoproteins that effectively operate as adversaries of host Fcγ receptors, inhibiting IgG-mediated immunity
- CMV has been associated with vasculopathy and neurocognitive impairment with associated immune markers including the accumulation of multifunctional terminally differentiated αβ T cells, γδ T cells, and NK cells.

7. PATHOGENESIS OF CRYPTOCOCCOSIS

- **Polysaccharide capsule.**
  - Cryptococcus spp. can block dendritic cell maturation by reducing MHC class II–dependent antigen presentation and inhibiting the production of IL-12 and IL-23.
  - Cryptococcus spp. can make large cells, called Titan cells that are greater than 12 µm and have a thickened cell wall.
  - Cryptococcus spp. also produce small (micro) cells of 2 to 4 µm that may be adapted for growth in macrophages.
- **Melanin production.**
- **Enzymes.**
  - Phospholipases degrade cell wall components and may aid tissue invasion.
  - Urease helps neutralize the reactive oxygen species and pH of the phagocytic cell.
- **Differential cellular response to phagocytes.**
  - A mechanism has been hypothesized to explain the outbreak of the C. gattii strain
  - In response to reactive oxygen species in the phagocyte, some cells stop growing and acquire an unusual morphology with tubularization of mitochondria, and other cells rapidly divide.

8. CLINICAL FEATURES OF HYDATID DISEASE:

- Infection in humans is usually asymptomatic, but large cysts in the liver can cause abdominal pain or obstruction, and pulmonary cysts can cause pain, cough, and hemoptysis.
- Caution is warranted if surgical removal of the cyst is considered, as anaphylaxis and/or dissemination of organisms can result from spillage of the cyst contents.

9. HHV8+ CONDITIONS

- Kaposi Sarcoma
- Primary effusion lymphoma
- Multicentric Castleman disease
- Germinotropic Lympho proliferative disorder
- KSHV Inflammatory Cytokine Syndrome

10. ONE LINERS:

- Babesia microti and Babesia divergens, are transmitted by ticks, **Ixodes scapularis** (deer tick) and **Ixodes ricinus** (sheep tick),
- Additional cases of babesiosis is due to **B. duncani** and **B. venatorum**.
- There are about 21 species of Leishmania that infect humans and about 30 species of sandflies that serve as vectors.
- **Elastases** and **trypsin-like proteases** facilitate invasion of host tissues in lymphatic filariasis.
- Multiplex PCR panels for detection of over 20 pathogens are now in clinical use for detection of pathogens in stool, respiratory samples, and cerebrospinal fluid, as a well as positive blood culture broths.
- High-throughput or next-generation sequencing methods, most often following PCR of the 16S rDNA gene for bacteria or the **ribosomal internal transcribed spacer (ITS)** region for fungi, allow rapid sequencing of large numbers of DNA molecules.
- Antiretroviral treatment of pregnant women with HIV infection and their newborn infants can reduce the transmission of HIV to offspring from 25% to less than 1%.
- **Brown and Brenn stain** are used to identify Bacillus anthracis

11. NEW TOPICS DISCUSSED:

- Pg 352 - Ebola Virus
- Pg 352 - Zika Virus
- Pg 353 - Dengue
- Pg 353 - Novel Coronavirus SARS-CoV-2 (COVID-19)
- Pg 373 - Non-tuberculous mycobacterial infections
- Pg 385 - Candida auris Infections
- Pg 386 - Pneumocystis Infections
- Pg 394 - Toxoplasmosis

Date Recognized	Infectious Agent	Manifestations
2007	Zika virus	Fever, rash, Guillain-Barré syndrome Fetal loss, microcephaly, neurologic complications in newborns
2014	Ebola virus	Fever, rash, hemorrhage, multisystem organ failure
2016	*Candida auris*	Candidemia, wound and ear infections resistant to multiple antifungal agents
2019	COVID-19	Fever, cough, dyspnea, pneumonia

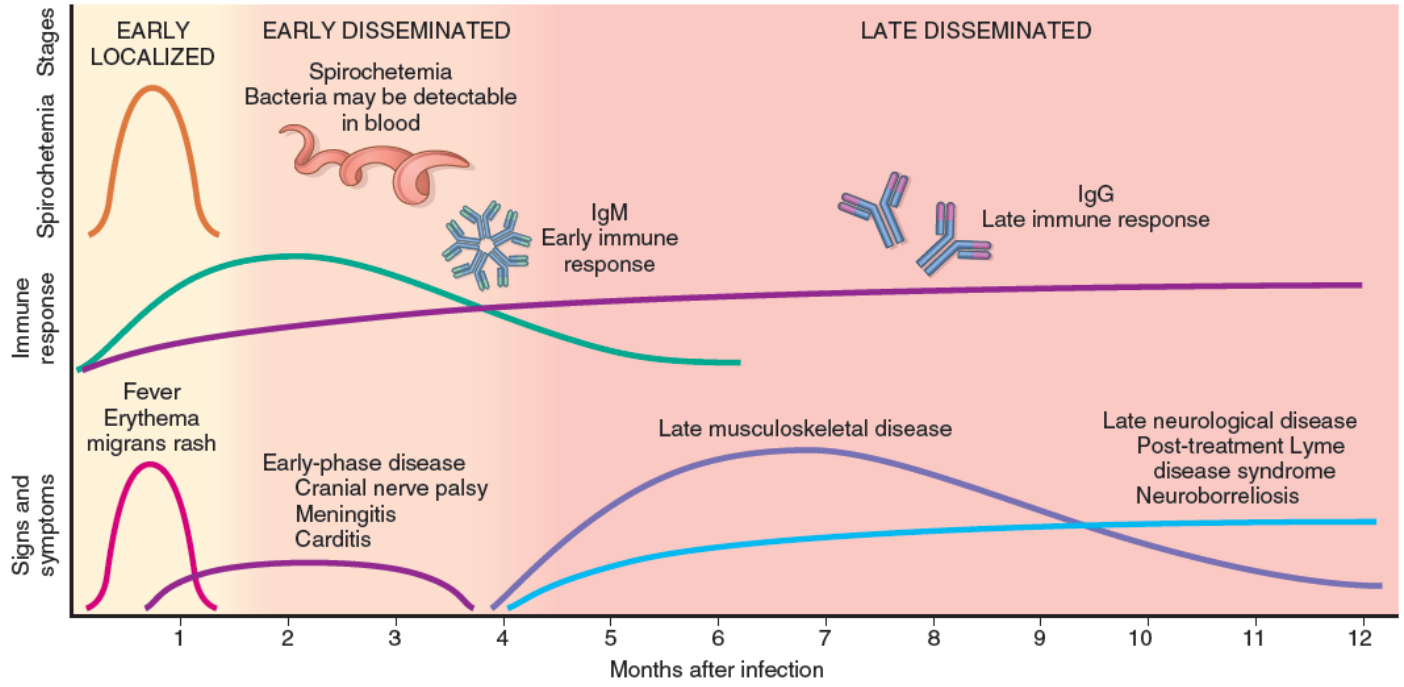


Figure 8.36 Lyme disease progresses through three clinically recognizable phases: early localized, early disseminated, and late disseminated. Although initial manifestations result directly from spirochete infection, later signs and symptoms are likely immune-mediated. (Figure modified from Dr. Charles Chiu, University of California, San Francisco, Calif. Used with permission.)

CLICK HERE FOR FEEDBACK